A NEW APPROACH TO IDENTIFY THE NATURE OF THE NOISE AFFECTING AN IMAGE.

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ABSTRACT

In this paper, we propose a method for the identification of the nature of noise and estimation of its statistical parameters, from only the degraded image. This then enables us to apply the appropriate filtering algorithm. The method is based on the criterion of maximum likelihood in order to detect homogeneous regions, therefore this enables to identify the nature of the noise. The identification criteria used are based on the analysis of local variations of the average and the standard deviation of the observed image. These statistical parameters are determined uniquely from regions of the image supposed to be homogeneous locally. After the nature of the noise has been identified, the statistical parameters of the noise are estimated.

1. INTRODUCTION

Filtering is one of the most important stages in the image pattern recognition chain, when the images are degraded. In fact the quality of the analysis heavily depends on the quality of the image.

The classical hypothesis of the image degraded by an additive noise is not always true. In fact, there is a significant class of images degraded either by a multiplicative noise which may or may not be dependant on the image or by an impulsive noise. We find these noises in various domains such as in industry, medicine, biology, radar or satellite images ...

Several methods of filtering an additive, multiplicative or impulsive noise have been developed [1-11]. But these methods suppose that the nature of the noise is known.

When one has no prior knowledge of the nature of the noise, one first has identify its nature and estimate its statistical characteristics (first and second moment).

2. IDENTIFYING THE NATURE OF THE NOISE

Let f(.,.) be the original image, degraded by an additive or multiplicative noise n(.,.) assumed white and independant of f(.,.) or by an impulsive noise; and g(.,.) the image observed. g(i,j) indicates the grey level at the point (i,j) with i,j = 0 to N. N is the number of pixels per line or per column of an image.

One notes E[X] and σ^2 [X] the average and the variance of the random variable X;

The estimation of a random variable X following a given criterion, will be noted $\hat{X}$.

2.1. The algorithm principle.

In order to identify the nature of the noise, each noise to be identified is characterised by its statistical parameters uniquely from the areas of the image supposed locally homogenous which allows us to define a decision criterion from the observations alone. Before clarifying this criterion we give the definition of the homogeneity of a region.

Definition of homogeneity :

Let F(.,.) be a window of the image g(.,.) with a size LxL (L=5,7) centered on the point (k,l). F(.,.) is divided into 9 different regions noted Ri (see fig.1); each region is made up of M pixels with M = Lx(L+1)/2 except the region M0 which only contains $(L-1)^2$ pixels. The central pixel belongs to all these regions.

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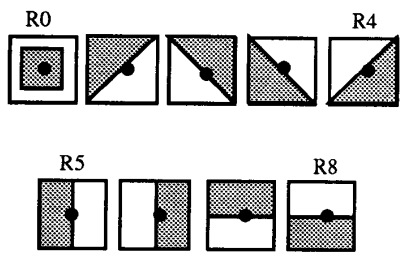

Fig. 1: Division of window into nine regions.

Assuming that the image is locally stationary, the notion of homogeneity therefore plays an important role in identifying the nature of the noise. That is to say that in the homogeneous regions only the statistics of the noise intervene. This means that the variance of a local homogeneous zone is weak compared to that of the noise.

2.1.1. Research of the homogeneous regions.

Among the regions set up we look for the most homogeneous region (the region Ri is said to be homogeneous if all its constituents stem from the same stochastic process). The detection of the most homogeneous regions is based on the criterion of maximum likelihood [1] [2] [12].
We affirm that the likelihood function (Fvs) is inversely proportional to :

$\sigma\,[g]^{-M}$ (and to $\sigma\,[g]^{-(L-2)^2}$ in the case of R0).

In this way the search for the maximum of the Fvs(.) (which will correspond to the most homogenous region Ri) is equal to that of the minimal variance (this is true for a noise supposed locally uniform or gaussian).

This paper limits itself to identifying three types of degradation : additive noise, multiplicative noise and impulsive noise.

2.1.2. Identification of the nature of the noise.

- **additive noise**

The equation of the observation is written: g= f+n; n(.,.) is a white centered noise not correlated to f. we therefore have :

$$\sigma^2\,[g]= \sigma^2\,[f] + \sigma^2\,[n]$$

We can prove that in a homogeneous region :

$$Var[g_h] \cong Var[n] \qquad (1)$$

g_h indicates an observation relative to a homogeneous region.

- **multiplicative noise**

g= f x n
n(.,.) has an average equal to one.
We have :

$$\sigma^2[g]= E\,[f^2]\,E[n^2] - E[f]^2\,E[n]^2$$

The homogenous regions are characterised by :
$E\,[f^2] = E[f]^2$
Finally we deduce :

$$\sigma\,[g]/\,E\,[g_h] = E[n\,] \text{ or } Var[g_h] \cong E^2[g_h]\,Var[n] \quad (2)$$

- **impulsive noise**

The observation equation is written as follows :

$$g = e.\,f + (1-e).\,z$$

Where e indicates a random binominal variable taking the value 1 with a probability p (absence of noise) and the value of zero with a probability of 1-p (noise present); z is a random variable which replaces the real grey level with a probability of 1-p. The random variables are independant of f and mutually independant.
This shows that for a homogeneous region one can write :

$$Var\,[g_h] \cong A(p)\,E^2\,[g_h] - B(p,z)\,E[g_h] + C(p,z). \quad (3)$$

With $A(p) = (1-p)/p$

$B(p,z) = 2(1-p)^2\,E[z]$
and

$$C(p,z)=p(1-p)^3.E^2[z]+(1-p).\{var[z]+pE^2[z]\} \qquad (4)$$

We deduce from equations (1) and (2) that the noise is additive if, and only if the variance in the homogeneous regions of the image being studied is constant, and multiplicative if, and only if, the variance ($var[g_h]$) according to the average ($E[g_h]$) of the homogeneous regions of the image is a parabola passing through the original.

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An image is affected by an impulsive noise if the variation of the variance ($var[g_h]$) according to the average ($E[g_h]$) in the homogeneous regions is a parabola but not passing through zero. In accordance with equations (3) and (4), identification of the coefficients of the parabola allows us to determine p, the average and the variance of z.

Figure 2 shows the appearance of the different curves which allows identifying the additive, multiplicative or impulsive nature of the noise.

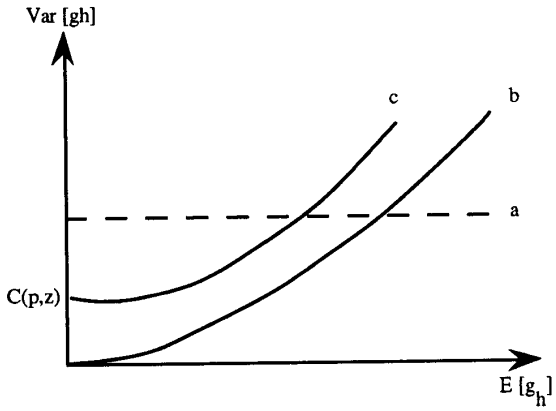

Figure 2 : Curves of Var $[g_h]$ according to E $[g_h]$.
a :additive noise. b : multiplicative noise.
c : impulsive noise.

2.2 Estimation of the standard deviation of an additive or multiplicative noise.

Once the nature of the noise is detected, in order to estimate the standard deviation of the noise we consider that in the homogeneous regions of the observed image, only the statisitics of the noise intervene. This signifies that the variance of a local homogeneous zone is weak in comparison with the variance of the noise. Estimating the value of the standard deviation of the noise is therefore equal to the average of the different values of the standard deviation of homogeneous regions in the image observed.

Let $\sigma\ [g_h]$ and $(\sigma\ [g_h]/E\ [g_h])$ be the random variables successively indicating the local standard deviation and the relationship between this last one and the local average, estimated from the locally homogeneous zones of the observed image.

In carrying out a lexographical scan of the image (line by line from top to bottom), the estimates of these two values fluctuate around their two average values $E[\sigma\ [g_h]]$ et $E[(\sigma[g_h]/\ E\ [g_h])]$, with factors of dispersion $\sigma[\sigma\ [g_h]]$ and $\sigma[(\sigma\ [g_h]/E\ [g_h])]$

If the test indicates an additive noise, its standard deviation will be :

$$\hat{\sigma} = \Sigma\ \sigma\ [g_h]\ /\ L$$

If not, the noise is multiplicative of the standard deviation :

$$\hat{\sigma} = \Sigma(\sigma\ [g_h]/\ E\ [g_h])/L$$

The sum concerns K homogeneous regions. If for each pixel of g(.,.) we look for the corresponding local homogeneous region, we will have L = NxN.

The results of indentification and estimation of the variance depend closely on the masks chosen in order to detect the local homogenous zones which allow the estimation of the parameters of the noise. We have compared the results obtained by using masks that we have proposed to those of Nagao [7]. The different results of the artificially degraded images show that the Nagao masks give the best results for a window with a size 7x7. On the other hand, for a window with a size 15x15, our masks give the most impressive results. The number of pixels with which the estimation is concerned, therefore plays an important role. Tables 1 and 2 show an example of an estimation of the standard deviation of the noise after the detection of its nature (additive or multiplicative). The results obtained have been tested on the artificially degraded image " woman with hat".

masks	nature of the detected noise	standard deviation of the estimated noise
Nagao	additive	09.21
Our masks	additive	12.22

Table 1 : results of the identification of the noise and the estimation of its standard deviation. The image "woman with hat" has been degraded by a centered gaussian additive noise of the stadnard deviation 12.

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masks	nature of the detected noise	standard deviation of the estimated noise
Nagao	multiplicative	0.15
Our masks	multiplicative	0.20

Table 2 : results of the identification of the noise and the estimation of its standard deviation. The image "woman with hat" has been degraded by a uniform multiplicative noise of the average 1 and the standard deviation 0.20.

These results show that our masks defined in a window with a size 15x15, give a better estimation of the standard deviation of the noise than those obtained by the Nagao masks defined in a window with a size 7x7.

3. CONCLUSION AND FUTURE WORK

The method we have developed has been tested on several images degraded by additive, multiplicative or impulsive noises. The results from the identification of the noise nature and the estimation of the obtained statistical parameters are satisfactory.

The method which we propose therefore constitutes an essential stage in the resolution of filtering problems when the nature of the noise is unknown. It also enables statistical parameters of noise to be estimated with precision.

The results obtained from the identification of the nature of the noise are thus encouraging but the calculation time remains relatively long. In fact, the decisional criteria which we have defined, are based on the one side, on the recording of the variations of a local standard deviation in comparison with the local average, and on the other side, on the interpretation of the different curves obtained. In order to reduce the calculation time, the new stage of our work will be devoted to the development of new criteria based on the different curves while avoiding the procedures of tracing them. Thus, we will optimize the performance of our method.

The work that we have presented, lies within the framework of automation of the filtering chain.

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